MIKE WONG

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SUMMARY

Fifth-year CS PhD candidate at Princeton building efficient, cross-stack AI infrastructure. Research focuses on optimizing inference systems by using domain-specific signals to filter out unnecessary information before expensive model execution occurs. Proven track record of designing and deploying these efficiency frameworks across web agents, live video analytics, and network security. Published at top venues (NSDI, SIGCOMM, CoNEXT). Graduating late 2026.

EDUCATION

Princeton University *Aug 2021 – Late 2026 (expected)*
Ph.D. in Computer Science. Advisor: Ravi Netravali. M.A. conferred Sep 2023.

New York University *Aug 2015 – Dec 2019*
B.A. in Computer Science, minor in Mathematics.

EXPERIENCE

Graduate Research Assistant, Princeton University *Aug 2021 – Present*
Advisor: Ravi Netravali

- Built **MadEye** (NSDI '24, first author), a camera-server system that dynamically filters and prioritizes spatial video data; cuts live inference streaming costs by **2–3.7**× while matching or improving baseline tracking accuracy.
- Lead author on position paper proposing pushing pipeline orchestration directly into network fabric using programmable NICs to reduce orchestration bottlenecks (arXiv 2502.15712).
- Co-authored system for detection-guided attention steering in vision-language models (in submission).

AI Systems Engineer Intern, BreezeML *May 2026 – Present*

- Design end-to-end distributed infrastructure to support high-throughput LLM agent evaluation workflows.
- Develop core pre-processing pipeline for unstructured knowledge bases and the scalable serving stack used for test generation and agent evaluation.

Research Intern, Cloud Systems Reliability Group, Microsoft Research *May 2025 – Aug 2025*

- Designed an execution stack for LLM web agents that uses capability-aware metrics to filter out redundant ReAct loops, shortcutting to direct actions to reduce inference cost and latency.
- First-author paper in preparation with Kevin Hsieh, Suman Nath, Ravi Netravali.

Research Intern, Networking Research Group, Microsoft Research *Jun 2022 – Aug 2022*

- Developed **NetVigil** (NSDI '24, co-first author), an anomaly detector that substitutes raw data-center packet traces with lightweight flow summaries, reducing telemetry monitoring overhead by **2.7–16.7**×.
- Led deployment over 16-VM cluster, outperforming SOTA baseline AUCs by 0.22. Resulted in a US patent.

Computer Scientist, National Security Agency *Feb 2020 – Aug 2021*

- Maintained and contributed core modules to **Ghidra**, the agency's open-source reverse-engineering framework.
- Built **Marvolo** (ECML PKDD '23, first author), an automated data augmentation engine that applies semantic filters to binaries, achieving up to 10% ML detection AUC improvements with a **79**× rewrite speedup.

Research Assistant, Rutgers University / NYU Courant *May 2019 – Jan 2021*
Advisors: Srinivas Narayana, Anirudh Sivaraman

- Co-developed **K2** (SIGCOMM '21), a stochastic program-synthesis eBPF compiler that filtered compilation noise to reduce kernel extension sizes by **6–26%** and accelerated equivalence checking by 5 orders of magnitude.
- Developed **Druzhba** (CoNEXT '20, first author), a programmable-switch hardware simulator, and co-developed **Chipmunk** (SIGCOMM '20), a synthesis-based switch compiler yielding a **51×** average speedup.

Computer Science Co-op, National Security Agency *Aug 2016 – Dec 2019*

- Programmed a multithreaded linear algebra library in C++ utilizing GMP and OpenMP for cryptographic workloads.
- Developed static analysis engines to scan and filter binary assembly listings for exploitable vulnerabilities.

SELECTED PUBLICATIONS

Full list at michaeldwong.github.io. * denotes equal contribution.

1. **MadEye: Boosting Live Video Analytics Accuracy with Adaptive Camera Configurations.** Mike Wong, Murali Ramanujam, Guha Balakrishnan, Ravi Netravali. USENIX NSDI 2024. [paper]
2. **NetVigil: Robust and Low-Cost Anomaly Detection for East-West Data Center Security.** Mike Wong*, Kevin Hsieh*, Santiago Segarra, Sathiya Kumaran Mani, et al. USENIX NSDI 2024. [paper]
3. **Marvolo: Programmatic Data Augmentation for ML-Driven Malware Detection.** Mike Wong, Edward Raff, James Holt, Ravi Netravali. ECML PKDD 2023. [paper]
4. **Synthesizing Safe and Efficient Kernel Extensions for Packet Processing.** Qiongwen Xu, Mike Wong, Tanvi Wagle, Srinivas Narayana, Anirudh Sivaraman. ACM SIGCOMM 2021. [paper]
5. **Testing Compilers for Programmable Switches Through Switch Hardware Simulation.** Mike Wong, Aatish Kishan Varma, Anirudh Sivaraman. ACM CoNEXT 2020. [paper]
6. **Switch Code Generation Using Program Synthesis.** Xiangyu Gao, Taegyun Kim, Mike Wong, et al. ACM SIGCOMM 2020. [paper]
7. **Capability-Aware Execution for Efficient Web Agents.** Mike Wong, Kevin Hsieh, Suman Nath, Ravi Netravali. *In preparation.*

TECHNICAL SKILLS

Languages	Python, C++, C, Rust, Java, JavaScript
ML / Systems	Distributed systems, model serving, infrastructure optimization, computer vision
LLMs / Agents	Agent evaluation stacks, knowledge base ETL, ReAct, prompt engineering, vLLM
Networking	eBPF, Verilog, Linux kernel, P4, programmable switches, packet processing
Tools	Ghidra, Git, Docker, Linux, LaTeX

PATENTS

Detecting Network Anomalies Using Network Flow Data. Tsuwang Hsieh, Santiago Martin Segarra, Sathiya Kumaran Mani, Srikanth Kandula, Mike Wong. US patent App. 18/367,775.

TEACHING & SERVICE

Instructor, Prison Teaching Initiative: MATH 020 Elementary Algebra (Fall 2025); MATH 101 Number Systems (Spring 2025).
Teaching Assistant, Princeton University: COS 316 (Systems Design, Fall 2023); COS 561 (Advanced Networks, Spring 2023).